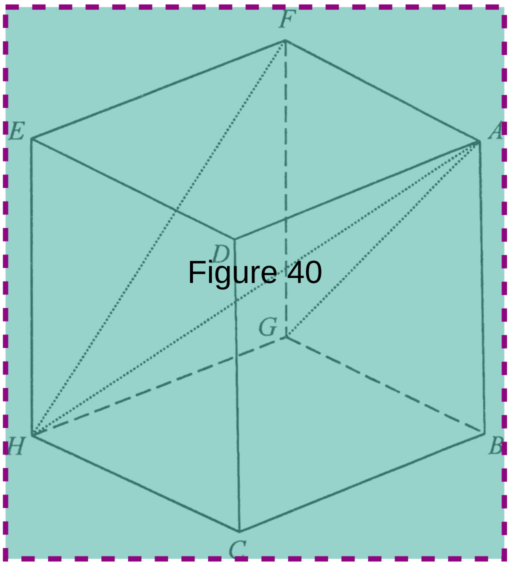
40. In the figure, $ABCDEFGH$ is a cube. Let α be the angle between ΔAFG and ΔAFH while β be the angle between ΔAFH and ΔFGH. Which of the following is true?



A. $\alpha < 60^\circ < \beta$

B. $\alpha < \beta < 60^\circ$

C. $60^\circ < \alpha < \beta$

D. $60^\circ < \beta < \alpha$

41. Let O be the origin. The coordinates of the points A and B are $(a, 0)$ and $(0, b)$ respectively, where a and b are positive numbers. If the circumcentre of ΔOAB lies on the straight line $4x + 16y = 17a$, then $a : b =$

A. $8 : 15$

B. $15 : 8$

C. $16 : 47$

D. $47 : 16$

42. If the first five digits and the last two digits of a seven-digit password are formed by a permutation of $1, 3, 5, 7, 9$ and a permutation of $2, 8$ respectively, how many different seven-digit passwords can be formed?

A. 10

B. 240

C. 480

D. 5 040